STATISTICS WORKSHEET-1

1. Bernoulli random variables take (only) the values 1 and 0.

a) True

2. Which of the following theorem states that the distribution of averages of iid variables, properly normalized, becomes that of a standard normal as the sample size increases?

a) Central Limit Theorem

3. Which of the following is incorrect with respect to use of Poisson distribution?

b) Modelling bounded count data

4. Point out the correct statement.

d) All of the mentioned

5. _____ random variables are used to model rates.

c) Poisson

6. Usually replacing the standard error by its estimated value does change the CLT.

b) False

7. 1. Which of the following testing is concerned with making decisions using data?

b) Hypothesis

8. Normalized data are centred at_____and have units equal to standard deviations of the original data.

a) 0

9. Which of the following statement is incorrect with respect to outliers?

c) Outliers cannot conform to the regression relationship

10. What do you understand by the term Normal Distribution?

The **normal (or Gaussian) distribution** is one particular kind of a bell-shaped curve. It is unimodal (that is, there is one peak"), symmetric (that is you can flip it around its mid-point) and its mean, median and mode are all equal. However, it is only one such distribution - others meet all those conditions and are not normal.

Many things are approximately normally distributed, for example the heights of adult human females or males, IQ, etc.

In addition, parametric statistics often assumes that something is normally distributed. E.g., ordinary least squares regression assumes the errors are normal.

11. How do you handle missing data? What imputation techniques do you recommend?

1**. Mean or Median Imputation**
When data is missing at random, we can use list-wise or pair-wise deletion of the missing observations. However, there can be multiple reasons why this may not be the most feasible option:

- There may not be enough observations with non-missing data to produce a reliable analysis

- In predictive analytics, missing data can prevent the predictions for those observations which have missing data
- External factors may require specific observations to be part of the analysis

In such cases, we impute values for missing data. A common technique is to use the mean or median of the non-missing observations. This can be useful in cases where the number of missing observations is low. However, for large number of missing values, using mean or median can result in loss of variation in data and it is better to use imputations.

2. **Multivariate Imputation by Chained Equations (MICE)**
MICE assumes that the missing data are Missing at Random (MAR). It imputes data on a variable-by-variable basis by specifying an imputation model per variable. MICE uses predictive mean matching (PMM) for continuous variables, logistic regressions for binary variables, bayesian polytomous regressions for factor variables, and proportional odds model for ordered variables to impute missing data.

3. Random Forest
Random forest is a non-parametric imputation method applicable to various variable types that works well with both data missing at random and not missing at random.
Random forest uses multiple decision trees to estimate missing values and outputs OOB (out of bag) imputation error estimates.

One caveat is that random forest works best with large datasets and using random forest on small datasets runs the risk of overfitting.
The extent of overfitting leading to inaccurate imputations will depend upon how closely the distribution for predictor variables for non-missing data resembles the distribution of predictor variables for missing data.

12. What is A/B testing?

A/B testing data science is a methodical way to evaluate the performance of two variants of a website, app, or campaign. It also goes by the name "split testing."

By dividing traffic into two groups and serving one group the A/B version while serving the other group the control, A/B testing seeks to determine what works and doesn't work for your business (the base version).

This enables us to evaluate the impact of various versions on conversion rates and response rates.

13. Is mean imputation of missing data acceptable practice?

The process of replacing null values in a data collection with the data's mean is known as **mean imputation**.

Mean imputation is typically considered terrible practice since it ignores feature correlation.

Consider the following scenario: we have a table with age and fitness scores, and an eight-year-old has a missing fitness

score. If we average the fitness scores of people between the ages of 15 and 80, the eighty-year-old will appear to have a significantly greater fitness level than he actually does.

Second, mean imputation decreases the variance of our data while increasing bias. As a result of the reduced variance, the model is less accurate and the confidence interval is narrower.

14. What is linear regression in statistics?

Linear regression is the practice of statistically calculating a straight line that demonstrates a relationship between two different items. linear regression is the simplest form of regression there is.

There are two characteristics that make that the case.

First, linear regressions are only capable of capturing linear relationships.

Secondly, linear regression is only capable of handling relationships between two variables.

15. What are the various branches of statistics?

Statistics may be divided into two main branches:

(1) Descriptive Statistics **(2)** Inferential Statistics

(1) Descriptive Statistics

Descriptive statistics deals with the collection of data, its presentation in various forms, such as tables, graphs and

diagrams and finding averages and other measures which would describe the data.

For example: Industrial statistics, population statistics, trade statistics, etc. Businessmen make use of descriptive statistics in presenting their annual reports, final accounts, and bank statements.

(2) Inferential Statistics

Inferential statistics deals with techniques used for the analysis of data, making estimates and drawing conclusions from limited information obtained through sampling and testing the reliability of the estimates.

For example: Suppose we want to have an idea about the percentage of the illiterate population of our country. We take a sample from the population and find the proportion of illiterate individuals in the sample. With the help of probability, this sample proportion enables us to make some inferences about the population proportion. This study belongs to inferential statistics.